

Suppose you have a simple dataset with two points: $[(x_1=1, y_1=1), (x_2=2, y_2=2)]$. We have $x_3=100$, and want to predict y_3 .

- a. Let's say you fit your KNN model with $k=1$, euclidean distance, and aggregator="mean" to the dataset. What will it predict for y_3 ? Why?

y_3 will predict 2. KNN model works by finding the k nearest neighbors to a given point. In this scenario, $k = 1$ so the nearest neighbor will be the point $x_2 = 2, y_2 = 2$. Since aggregator = "mean", the prediction for y_3 will be the average of the target values which equal to 2.

- b. Let's say you fit your linear regression model (i.e., degree=1) to this data. What will it predict for y_3 ? Why?

y_3 will predict 100. A linear regression model with degree = 1 fits a straight line to the data that minimizes the sum of the squared differences. In this case, the line would be $y=x$, and the prediction for y_3 would simply be $y_3 = x_3 = 100$.

- c. Based on your answers to a. and b., what is the *inductive bias* of both models? By inductive bias, we are referring to the assumptions are built into each model about how to generalize to new data that's unlike data it has seen before. How would you describe these assumptions for each model?

The inductive bias of the KNN model is that the target value of a new point is based on the mean or mode of the target values of its nearest neighbors. On the other hand, the inductive bias of the linear regression model is that the relationship between the input and target variables can be described by a simple straight line. The inductive bias of the KNN model is based on the assumption of local similarity, while the inductive bias of the linear regression model is based on the assumption of a linear relationship between the input and target variables.